AUDIT REPORT

ddmrp-buffers: demo video audit report

A slide-by-slide record of every number, assumption and calculation in the demo video, with its source.

Item	Value
Repository	https://github.com/kavya2693/ddmrp-buffers
Numbers taken from commit	`ef2e4977bbb5278cf24e612031c686ff3d296161` (outputs last changed in `ef2e497 (2026-10-13 17:53:42 +0400)`)
Video	`docs/demo/demo.mp4` · 86.4 s · 1920x1080 · 30 fps · silent
Generated	2026-10-14 14:20
Automated checks	54 pass · 0 fail · 2 manual (of 56 figures)
Assumptions	14 (A-01 to A-14)

How to use this report

Every number or claim on screen has an ID (F-01, F-02, ...) and every modelling assumption has an ID (A-01, ...). Cite the ID. Each figure lists its source file, the exact column, row or key, the formula for derived numbers, and an automated check result. The change-impact matrix (section 5) shows which slides and files change if an input or assumption changes.

Types: direct = read straight from a repo output; derived = computed from direct values (formula given); illustrative = drawing or example, not data; general knowledge = context not produced by the repo.

Regenerate

```
make run                                    # data/*.csv, outputs/results.json, frontier.csv, segments.csv, 3 PNGs (~20 s)
python3 docs/demo/source/tools/darkmap.py outputs docs/demo/source/assets
PYTHONPATH=src .venv/bin/python docs/demo/source/tools/export_po_sheet.py docs/demo/source/assets/po_sheet.json 2025-09-08
python3 docs/demo/source/tools/inject_sheet.py docs/demo/source/assets/po_sheet.json docs/demo/source/compositions/value.html --rows LUB-0022,LUB-0304,LUB-0540,LUB-0079,LUB-0788,LUB-0035 --animate LUB-0022,LUB-0304,LUB-0079,LUB-0035
cd docs/demo/source && npx --yes hyperframes@0.8.156 check && npx --yes hyperframes@0.8.156 render -f 30 -q looks -o renders/ddmrp-buffers-master.mp4 && cd -
ffmpeg -y -i docs/demo/source/renders/ddmrp-buffers-master.mp4 -c:v libx264 -preset slow -crf 25 -pix_fmt yuv420p -movflags +faststart -an docs/demo/demo.mp4
python3 docs/demo/source/tools/build_audit_report.py docs/demo/source/audit/facts.yaml   # stills, checks, MD/DOCX/PDF
```

1. Figures

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
F-01	3 industry chips	01	author's framing	-	-	general knowledge	manual: Context statement, not a repo output.
F-02	4 problem cards	01	README 'The problem'	README lines 13-27	-	general knowledge	✓ verified
F-03	800 SKUs	02, 03, 05, 08, 10	outputs/results.json	dataset.skus	-	direct	✓ verified
F-04	21.5M in stock	02	outputs/results.json	opening_stock_value = 21,516,738	/ 1e6, 1 decimal	direct	✓ verified
F-05	172 days of cover	02	outputs/results.json	opening_dio_days = 172.03	-	direct	✓ verified
F-06	26% imports	02	outputs/results.json	dataset.import_share = 0.26375	x 100	direct	✓ verified
F-07	45–90 d lead	02	data/skus.csv	lead_time_days where source = import (min, max)	-	direct	✓ verified
F-08	4,702 bulk orders	02, 05	outputs/results.json; data/bulk_orders.csv	dataset.bulk_orders	-	direct	✓ verified
F-09	3–21 d notice / 3 to 21 days before they ship	02, 04, 05	data/bulk_orders.csv	due_date - placed_date (min, max)	-	direct	✓ verified
F-10	60% of SKU-days sell nothing	02	outputs/results.json	dataset.zero_demand_day_share = 0.5968	x 100, whole number	direct	✓ verified
F-11	84-day average / ADU 84 d / 84-d window	03, 05	src/ddmrp_buffers/config.py:24	adu_window_days = 84	-	direct	✓ verified
F-12	EVERY 7 DAYS / weekly	03, 05	src/ddmrp_buffers/config.py:25; policies.py:154	recalc_every_days = 7	-	direct	✓ verified
F-13	EVERY DAY / all 800 SKUs, daily / per-day call	03, 05	src/ddmrp_buffers/simulate.py:77-88	for t in range(n_days): ... policy.order(...)	-	direct	✓ verified
F-14	47% less stock / same fill rate / case + MOQ	03	outputs/results.json; src/ddmrp_buffers/buffers.py:104-106	matched_service DDMRP vs legacy year-2 DIO; round_up	(1 - 52.14 / 97.64) x 100 = 46.6% -> 47%; DIO read at the legacy fill rate (A-08)	derived	✓ verified
F-15	on hand + on order – orders already placed	04	src/ddmrp_buffers/buffers.py:99-101; policies.py:160-165	net_flow()	-	direct	✓ verified
F-16	red 105 / yellow 205 / green 275 / net flow 170 -> 140 / Order 144	04	tests/test_buffers.py:12-27 (test_zones_match_hand_worked_example)	assertions in the test	275 - 140 = 135 -> 144 (cases of 12)	direct	✓ verified
F-17	Without the customer order: 108.	04	tests/test_buffers.py:12-27 inputs; buffers.order_quantity / round_up	net flow 170 (120 + 50), TOG 275, MOQ 50, case 12	275 - 170 = 105 -> ceil(105 / 12) x 12 = 108	derived	✓ verified
F-18	94.49% vs 94.51%	04, 09	outputs/results.json	policies['DDMRP, no order-spike visibility'] / ['reorder point (tuned)'].year_2.value_fill_rate	x 100, 2 decimals	direct	✓ verified
F-19	seed 20240101	05	src/ddmrp_buffers/config.py:8	SEED	-	direct	✓ verified
F-20	1,096 d × 800 / 2023: 4.7% missing	05	outputs/results.json	dataset.days_total, dataset.history_missing_share	-	direct	✓ verified
F-21	+14–35 d holds	05	src/ddmrp_buffers/simulate.py:49-50	10% of import orders held rng.integers(14, 36) days	-	direct	✓ verified
F-22	ADU 84 d · LTF · VF / red · yellow · green / ≤ TOY? yes: up to TOG	05	src/ddmrp_buffers/buffers.py:35-119	average_daily_usage, lead_time_factor, variability_factor, zones, net_flow, order_quantity	-	direct	✓ verified
F-23	4 classes · 180 d	05	src/ddmrp_buffers/classify.py:13-14; config.py:66	RUNNER, REPEATER, STRANGER, DORMANT; window_days = 180	-	direct	✓ verified
F-24	day 731 / 731 · 731 simulated days	05, 08	outputs/results.json	dataset.days_simulated	-	direct	✓ verified
F-25	11 z · 8 red sizes	05	src/ddmrp_buffers/cli.py:14-15	len(ROP_Z_SWEEP), len(DDMRP_RED_SWEEP)	-	direct	✓ verified
F-26	+ 3 fresh ranges / 4 ranges	05, 08	outputs/results.json robustness; cli.py:252	len(robustness) = 4 (main + 3)	-	direct	✓ verified
F-27	×90 / 90 runs	05	src/ddmrp_buffers/cli.py:18-25, 76-93, 120-131, 175-177	5 arms + 19 sweep runs on the main range; 3 policies + 19 sweep runs on each of 3 extra ranges	5 + 19 + 3 x (3 + 19) = 90	derived	✓ verified
F-28	≈ 20 s / ~20 s	05, 10	README 'Running it'	'(~20 s)'	-	direct	✓ verified
F-29	98.46%	05, 06,	outputs/results.json	matched_service.target_	x 100	direct	✓ verified

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
		08		value_fill_rate_year_2			
F-30	generate_data.py, data.load(), context(), simulate.py, buffers.py, classify.py, policies.py, config.py, Ddmrp.order(), metrics.kpis(), metrics.py, cli.py, charts.py, outputs/	05	src/ddmrp_buffers/, scripts/	files + functions	-	direct	✓ verified
F-31	LOCAL BLEND · 5-DAY LEAD / IMPORT · 79-DAY LEAD	06	outputs/buffer_chart.png (panel titles); data/skus.csv	sku, source, lead_time_days	-	direct	✓ verified
F-32	legacy fill 98.46% + three rings	06	outputs/frontier.png (axes); compositions/s04-routes.html (X/Y mapping)	tick positions in the dark image	x = 332 + (fill - 90) x 150.1, y = 968 - (DIO - 30) x 13.514 (image pixels)	illustrative	manual: Annotation placed by the chart's own axis ticks; the curves are the repo's PNG, not redrawn. Visually confirmed: the rings sit on the legacy dot and on both swept curves.
F-33	97.6 / 65.9 / 52.1	06	outputs/results.json	policies['min-max (legacy)'].year_2.dio_days; matched_service.dio_at_target	-	direct	✓ verified
F-34	–47%	06, 08	outputs/results.json	F-33	(52.14 / 97.64 - 1) x 100 = -46.6%	derived	✓ verified
F-35	MON 2025-09-08 · DDMRP	07	demo assets/po_sheet.json (re-run of the repo's default DDMRP simulation)	day, weekday	-	direct	✓ verified
F-36	49 purchase orders	07	demo assets/po_sheet.json	orders = number of SKUs with order qty > 0	-	direct	✓ verified
F-37	127,376 at cost	07	demo assets/po_sheet.json	order_value = sum(order qty x unit cost)	-	direct	✓ verified
F-38	7 pulled in by customer orders	07	demo assets/po_sheet.json	with_customer_orders = 7; triggered_by_customer_orders = 7 (net flow without them would sit above top of yellow)	-	direct	✓ verified
F-39	+ 43 more SKUs to order today	07	F-36	49 - 6 rows	49 - 6 = 43	derived	✓ verified
F-40	LUB-0022, LUB-0304, LUB-0540, LUB-0079, LUB-0788, LUB-0035	07	demo assets/po_sheet.json, injected into compositions/value.html	rows by SKU	due = order day + nominal lead time; rows ordered most urgent first (net flow / top of green); LUB-0540 kept as a high-volume line	direct	✓ verified
F-41	local · 5–14 days / imports · 45–90 days	07	data/skus.csv	lead_time_days by source (min, max)	-	direct	✓ verified
F-42	+3 / +6 / +76 / +42 with green checks	07	demo assets/po_sheet.json	net_flow + order_qty >= tog	-	direct	✓ verified
F-43	–13.7 d (65.9 → 52.1 days)	08	outputs/results.json	matched_service.dio_at_target	65.888 - 52.140 = 13.75 -> 13.7	derived	✓ verified
F-44	59.6 d	08	outputs/results.json	max(robustness[].ddmrp_dio_at_legacy_fill)	max over 4 ranges	direct	✓ verified
F-45	legacy: 97.6–108.2 days	08	outputs/results.json	robustness[].legacy_dio (min, max)	-	direct	✓ verified
F-46	5.1×	08	outputs/results.json	policies[*].year_2.purchase_orders_per_month: 1,004.1 / 195.1	1,004.1 / 195.1 = 5.15	derived	✓ verified
F-47	195 → 1,004 a month	08	outputs/results.json	policies[*].year_2.purchase_orders_per_month	-	direct	✓ verified
F-48	5 policies	08, 10	outputs/results.json	len(policies)	-	direct	✓ verified
F-49	96.61% · 39.9 d	09	outputs/results.json	policies.DDMRP.year_2	-	direct	✓ verified
F-50	94.49% · 39.8 d	09	outputs/results.json	policies['DDMRP, no order-spike visibility'].year_2	-	direct	✓ verified
F-51	94.51% · 40.1 d	09	outputs/results.json; config.py:59	policies['reorder point (tuned)'].year_2	-	direct	✓ verified
F-52	13.6% of demand value	09	outputs/results.json	dataset.bulk_share_of_simulated_demand_value = 0.13556	x 100	direct	✓ verified
F-53	about 70%	09	outputs/results.json	legacy DIO, matched-service DIO of reorder point and DDMRP	(97.64 - 65.89) / (97.64 - 52.14) = 0.698	derived	✓ verified
F-54	MIT	10	LICENSE line 1	-	-	direct	✓ verified
F-55	Python 3.11 · NumPy · pandas · matplotlib	10	pyproject.toml	requires-python, dependencies	-	direct	✓ verified

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
F-56	$ make run	10	Makefile	run target	-	direct	✓ verified

2. Automated check run (2026-10-14 14:20)

ID	Expected on slide	Recomputed from repo	Result
F-01	3 industry chips	-	manual
F-02	true	true	✓ pass
F-03	800	800	✓ pass
F-04	21.5M	21.5M	✓ pass
F-05	172	172	✓ pass
F-06	26%	26%	✓ pass
F-07	45–90	45–90	✓ pass
F-08	4,702	4,702	✓ pass
F-09	3–21	3–21	✓ pass
F-10	60%	60%	✓ pass
F-11	true	true	✓ pass
F-12	true	true	✓ pass
F-13	true	true	✓ pass
F-14	true	true	✓ pass
F-15	true	true	✓ pass
F-16	true	true	✓ pass
F-17	108	108	✓ pass
F-18	94.49% vs 94.51%	94.49% vs 94.51%	✓ pass
F-19	true	true	✓ pass
F-20	1,096 d, 4.7%	1,096 d, 4.7%	✓ pass
F-21	true	true	✓ pass
F-22	true	true	✓ pass
F-23	true	true	✓ pass
F-24	731	731	✓ pass
F-25	11 z, 8 red	11 z, 8 red	✓ pass
F-26	true	true	✓ pass
F-27	90	90	✓ pass
F-28	true	true	✓ pass
F-29	98.46%	98.46%	✓ pass
F-30	true	true	✓ pass
F-31	true	true	✓ pass
F-32	legacy fill 98.46% + three rings	-	manual
F-33	97.6 / 65.9 / 52.1	97.6 / 65.9 / 52.1	✓ pass
F-34	-47%	-47%	✓ pass
F-35	true	true	✓ pass
F-36	49	49	✓ pass
F-37	127,376	127,376	✓ pass
F-38	true	true	✓ pass
F-39	43	43	✓ pass
F-40	true	true	✓ pass
F-41	5–14 / 45–90	5–14 / 45–90	✓ pass
F-42	true	true	✓ pass
F-43	-13.7	-13.7	✓ pass
F-44	59.6	59.6	✓ pass
F-45	97.6–108.2	97.6–108.2	✓ pass
F-46	5.1	5.1	✓ pass
F-47	195 → 1,004	195 → 1,004	✓ pass
F-48	5	5	✓ pass
F-49	96.61% · 39.9 d	96.61% · 39.9 d	✓ pass
F-50	94.49% · 39.8 d	94.49% · 39.8 d	✓ pass
F-51	94.51% · 40.1 d	94.51% · 40.1 d	✓ pass
F-52	13.6%	13.6%	✓ pass
F-53	70%	70%	✓ pass
F-54	true	true	✓ pass
F-55	true	true	✓ pass
F-56	true	true	✓ pass

3. Assumptions register

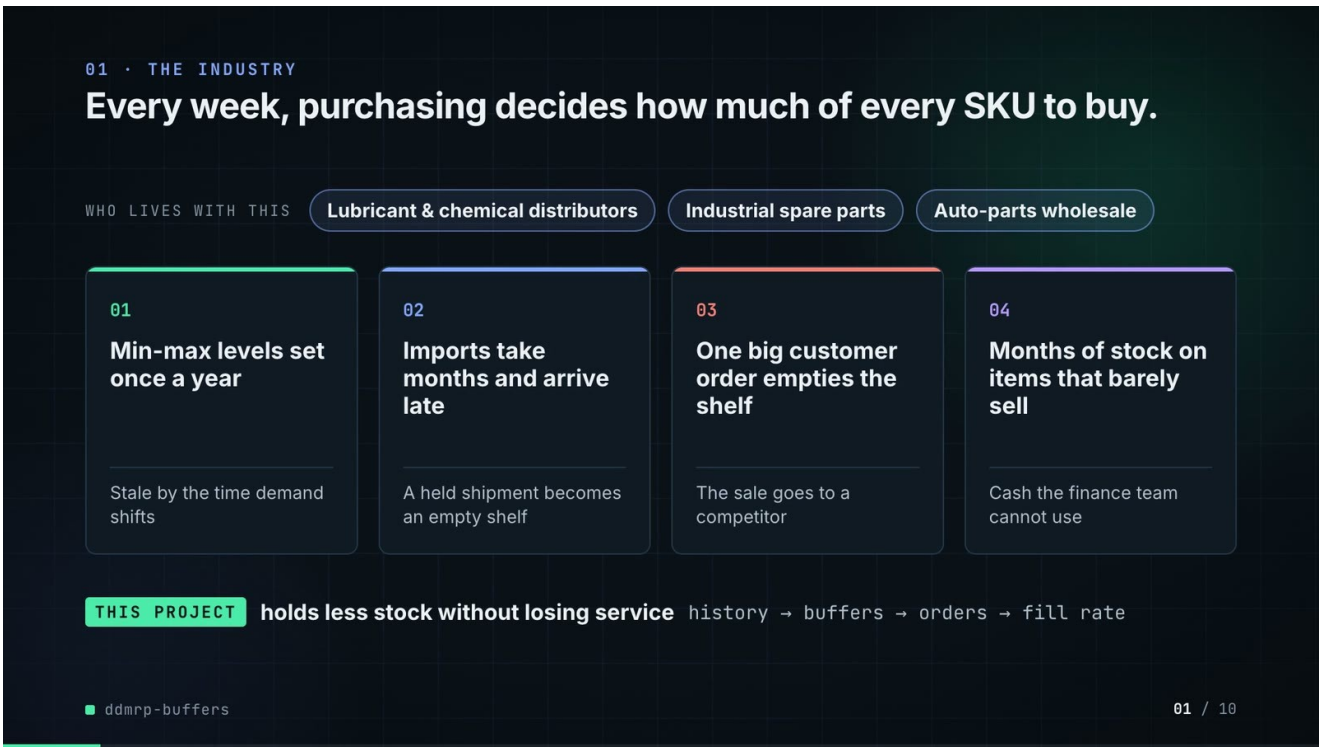
ID	Assumption	Source	Figures affected	Slides affected
A-01	All data is synthetic, produced by a seeded generator sized to the production problem (800 SKUs, three years of daily demand, 4,702 bulk orders). The production data is proprietary.	scripts/generate_data.py; src/ddmrp_buffers/data.py:95-188 (generate); config.py:8 (SEED = 20240101); README 'The data'	all figures	slides 01, 02, 03, 04, 05, 06, 07, 08, 09, 10
A-02	Bulk fleet orders are announced 3 to 21 days before they ship: 60% of SKUs packed in 18 L or more get 2-8	src/ddmrp_buffers/data.py:154-171; README 'Known limitations' (the biggest gap)	F-08, F-09, F-16, F-17, F-18, F-36, F-37, F-38, F-49, F-50,	slides 02, 04, 05, 07, 09

ID	Assumption	Source	Figures affected	Slides affected
	such orders a year (15% of the rest get 2). Their share (13.6% of demand value) and notice period are generator choices, not measurements; DDMRP's edge over the reorder point depends on them.		F-51, F-52	
A-03	Unfilled demand is lost, not backordered (a workshop buys the drum elsewhere). Fill rate is the share of demand value shipped on the day it was due.	src/ddmrp_buffers/simulate.py:81-83; metrics.py:27; docs/adr/0001-compare-at-matched-service.md	F-18, F-29, F-33, F-34, F-43, F-44, F-49, F-50, F-51, F-53	slides 04, 05, 06, 08, 09
A-04	Lead times: local blends 5-14 days, imports 45-90 days. Realised lead times drift (sd 15% blends, 10% imports) and 10% of import orders are held at the port for 14-35 extra days.	src/ddmrp_buffers/data.py:109-114; simulate.py:42-51 (actual_lead_times)	F-07, F-21, F-33, F-34, F-41, F-44, F-49, F-50, F-51	slides 02, 05, 06, 07, 08, 09
A-05	DDMRP settings: 84-day ADU window, buffers recalculated every 7 days, lead-time factor 0.7 / 0.5 / 0.3 by lead-time band, variability factor 0.3 / 0.5 / 0.8 by weekly CV, a customer order counts early if it is at least half the red zone and due within the lead time, order cycle 7 days (blend) / 30 days (import).	src/ddmrp_buffers/config.py:22-41 (DdmrpSettings)	F-11, F-12, F-22, F-33, F-34, F-36, F-37, F-38, F-40, F-42, F-43, F-44, F-49, F-50	slides 03, 05, 06, 07, 08, 09
A-06	Legacy min-max: min = usage over lead time + 30 days, max = min + 90 days (blend) or 180 days (import), reset once a year from last year's average daily usage.	src/ddmrp_buffers/config.py:44-50; policies.py:60-79	F-29, F-33, F-34, F-45, F-47, F-53	slides 05, 06, 08, 09
A-07	Tuned reorder point: z = 1.88 (97% cycle service), same 84-day window, weekly recalculation and order cycle as DDMRP. The safety knobs are swept over 11 z values (0.84-5.5) and 8 DDMRP red-zone multipliers (0.5-4.0).	src/ddmrp_buffers/config.py:53-61; cli.py:14-15 (ROP_Z_SWEEP, DDMRP_RED_SWEEP)	F-25, F-27, F-33, F-34, F-43, F-44, F-51, F-53	slides 05, 06, 08, 09
A-08	Stock at matched service = DIO linearly interpolated along each policy's swept (fill rate, DIO) curve at the legacy policy's year-2 fill rate (98.46%).	src/ddmrp_buffers/metrics.py:42-49; cli.py:172-174; docs/adr/0001-compare-at-matched-service.md	F-14, F-29, F-33, F-34, F-43, F-44, F-53	slides 03, 05, 06, 08, 09
A-09	Headline window is year 2 of the simulation (2025), after the opening overstock has worked through; 2023 is history only, 2024-2025 are simulated (731 days).	src/ddmrp_buffers/cli.py:141-143; config.py:10-12	F-24, F-29, F-33, F-34, F-43, F-44, F-45, F-46, F-47, F-49, F-50, F-51, F-53	slides 05, 06, 08, 09
A-10	Opening stock reflects generous cover (lognormal around 110 days, x1.6 on imports and on declining or phased-out SKUs): 21.5 M at cost, 172 days of trailing usage.	src/ddmrp_buffers/data.py:179-186; metrics.py:35-39	F-04, F-05	slides 02
A-11	Lead-time draws use simulation seed 7 for the main range; the three extra ranges use their own seed for both the range and the draws. Slide 07 shows one day (Monday 2025-09-08) of the same DDMRP run, chosen by the video author as a weekday with both imports and customer-order pulls.	src/ddmrp_buffers/simulate.py:55 (seed=7); cli.py:120-131; demo tools/export_po_sheet.py	F-35, F-36, F-37, F-38, F-39, F-40, F-42, F-44, F-45	slides 07, 08
A-12	Purchase orders are counted, not costed: there is no ordering or freight cost in the comparison.	src/ddmrp_buffers/metrics.py:31; README 'Known limitations'	F-46, F-47	slides 08
A-13	Video rounding: days to 1 decimal, fill rates to 2 decimals, headline percentages to whole numbers, money to 0.1 M or whole units; '~20 s', 'about 70%' and '60%' are rounded labels.	this audit (display convention)	F-04, F-05, F-06, F-10, F-14, F-20, F-34, F-43, F-46, F-52, F-53	slides 02, 03, 05, 06, 08, 09
A-14	Slide 01 (industries, everyday problems) is general distribution knowledge used as context. It is not a result of this repo and carries no numbers.	author's framing; README 'The problem' supports all four problems	F-01, F-02	slides 01

4. Slide by slide

Slide 01: The industry

0.0-7.2 s · composition `compositions/s01-context.html`



On-screen text (verbatim)

01 · THE INDUSTRY

Every week, purchasing decides how much of every SKU to buy.

WHO LIVES WITH THIS · Lubricant & chemical distributors · Industrial spare parts · Auto-parts wholesale

01 Min-max levels set once a year — Stale by the time demand shifts

02 Imports take months and arrive late — A held shipment becomes an empty shelf

03 One big customer order empties the shelf — The sale goes to a competitor

04 Months of stock on items that barely sell — Cash the finance team cannot use

THIS PROJECT holds less stock without losing service · history → buffers → orders → fill rate

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-01	3 industry chips	`author's framing` · -	-	manual
F-02	4 problem cards	`README 'The problem'` · README lines 13-27	-	✓

Assumptions: A-14

Illustrative, not data:

- The whole slide is context; no repo figures.

To change this slide: Edit the chips (#ctx-c1..3) and cards (#ctx-k1..4) in compositions/s01-context.html.

Slide 02: This project

6.8-11.6 s · composition `compositions/s02-hook.html`



On-screen text (verbatim)

02 · THIS PROJECT

800 SKUs, 21.5M in stock. 172 days of cover. Cut it without losing customers.

26% imports, 45–90 d lead · 4,702 bulk orders, 3–21 d notice · 60% of SKU-days sell nothing

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	800 SKUs	`outputs/results.json` · dataset.skus	-	✓
F-04	21.5M in stock	`outputs/results.json` · opening_stock_value = 21,516,738	/ 1e6, 1 decimal	✓
F-05	172 days of cover	`outputs/results.json` · opening_dio_days = 172.03	-	✓
F-06	26% imports	`outputs/results.json` · dataset.import_share = 0.26375	x 100	✓
F-07	45–90 d lead	`data/skus.csv` · lead_time_days where source = import (min, max)	-	✓
F-08	4,702 bulk orders	`outputs/results.json; data/bulk_orders.csv` · dataset.bulk_orders	-	✓
F-09	3–21 d notice / 3 to 21 days before they ship	`data/bulk_orders.csv` · due_date - placed_date (min, max)	-	✓
F-10	60% of SKU-days sell nothing	`outputs/results.json` · dataset.zero_demand_day_share = 0.5968	x 100, whole number	✓

Assumptions: A-01, A-02, A-04, A-10

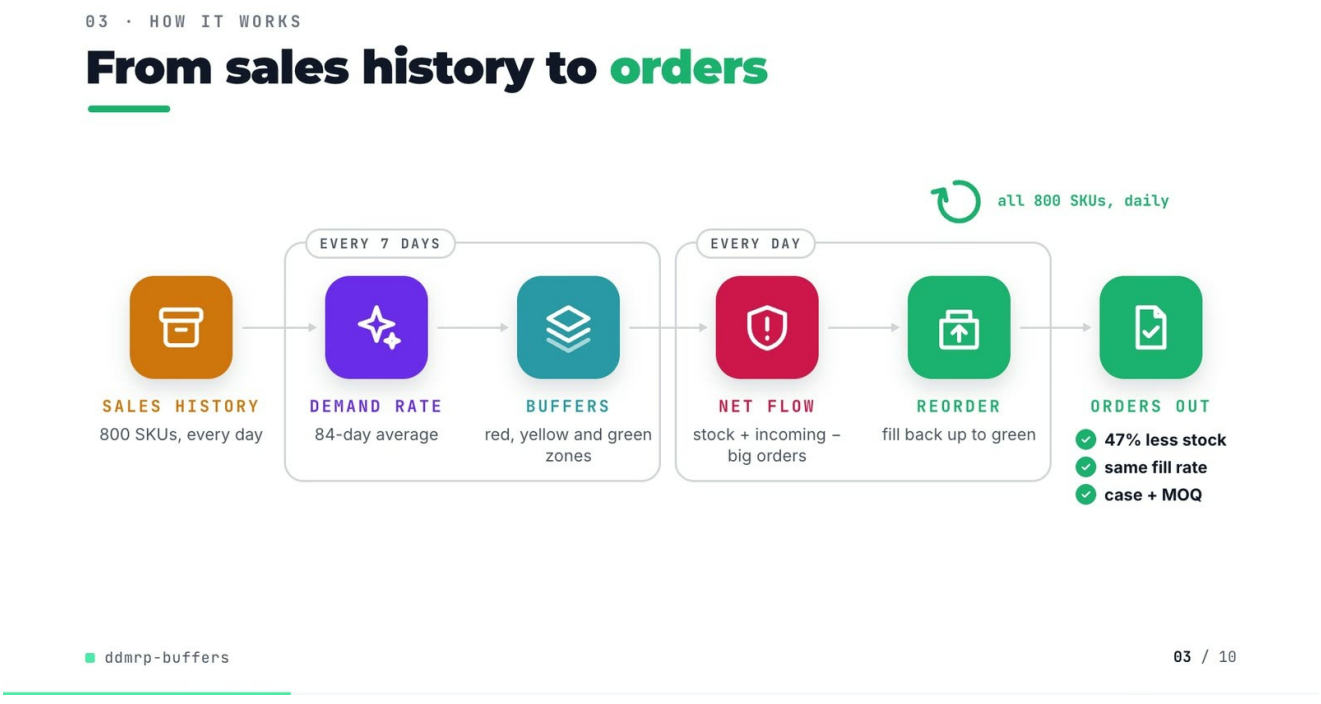
Illustrative, not data:

- Rings in the background are decoration.

To change this slide: Headline spans #hook-l1..l3 and chips #hook-f1..f3 in compositions/s02-hook.html; numbers from outputs/results.json (dataset, opening_*).

Slide 03: How it works (white chapter, slide A)

11.3-19.3 s · composition `compositions/explainer.html (CONFIG.blocks, CONFIG.groups, CONFIG.loop)`



On-screen text (verbatim)

03 · HOW IT WORKS

From sales history to orders

SALES HISTORY 800 SKUs, every day → [EVERY 7 DAYS: DEMAND RATE 84-day average → BUFFERS red, yellow and green zones] → [EVERY DAY: NET FLOW stock + incoming − big orders → REORDER fill back up to green] → ORDERS OUT ✓ 47% less stock ✓ same fill rate ✓ case + MOQ

all 800 SKUs, daily (loop over REORDER)

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	800 SKUs	`outputs/results.json` · dataset.skus	-	✓
F-11	84-day average / ADU 84 d / 84-d window	`src/ddmrp_buffers/config.py:24` · adu_window_days = 84	-	✓
F-12	EVERY 7 DAYS / weekly	`src/ddmrp_buffers/config.py:25`; `policies.py:154` · recalc_every_days = 7	-	✓
F-13	EVERY DAY / all 800 SKUs, daily / per-day call	`src/ddmrp_buffers/simulate.py:77-88` · for t in range(n_days): ... policy.order(...)	-	✓
F-14	47% less stock / same fill rate / case + MOQ	`outputs/results.json`; `src/ddmrp_buffers/buffers.py:104-106` · matched_service DDMRP vs legacy year-2 DIO; round_up	(1 - 52.14 / 97.64) x 100 = 46.6% -> 47%; DIO read at the legacy fill rate (A-08)	✓

Assumptions: A-05, A-08

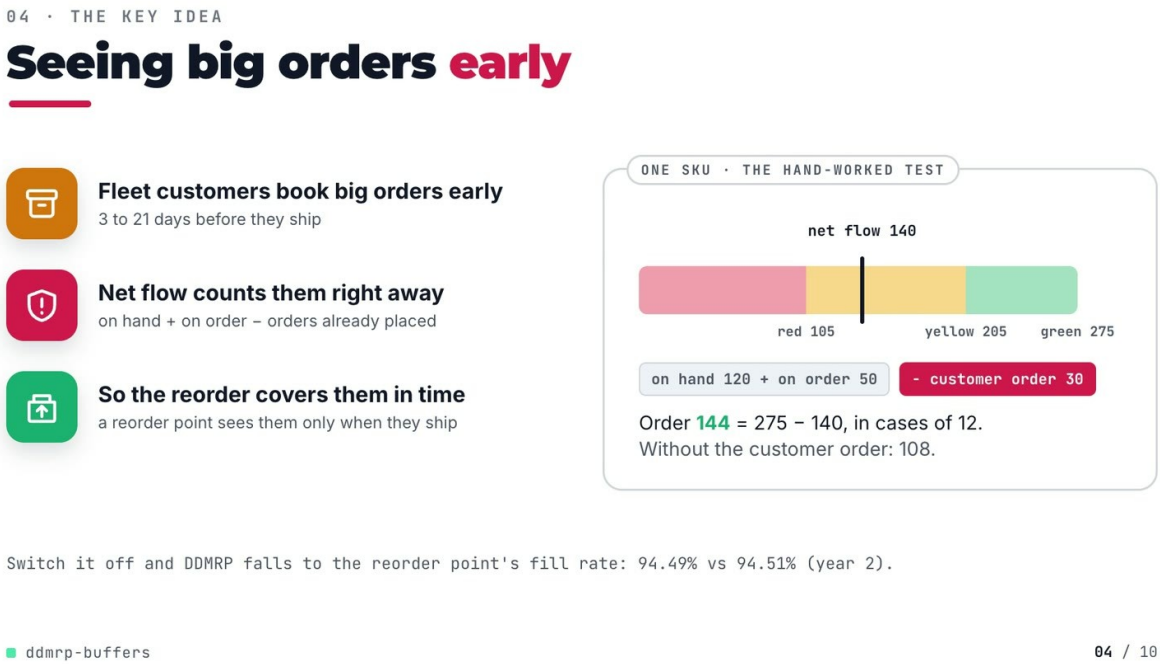
Illustrative, not data:

- Icons and colours are a visual language, not data.
- 'same fill rate' and '47% less stock' hold at matched service (A-08), not at default settings.

To change this slide: Edit CONFIG in compositions/explainer.html; badge truth from outputs/results.json matched_service (F-14).

Slide 04: The key idea (white chapter, slide B)

19.3-26.3 s · composition `compositions/explainer.html` (slide B markup + its timeline block)



On-screen text (verbatim)

04 · THE KEY IDEA

Seeing big orders early

Fleet customers book big orders early — 3 to 21 days before they ship

Net flow counts them right away — on hand + on order − orders already placed

So the reorder covers them in time — a reorder point sees them only when they ship

ONE SKU · THE HAND-WORKED TEST: net flow 170 → 140; red 105 · yellow 205 · green 275; on hand 120 + on order 50; − customer order 30

Order 144 = 275 − 140, in cases of 12. Without the customer order: 108.

Switch it off and DDMRP falls to the reorder point's fill rate: 94.49% vs 94.51% (year 2).

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-09	3–21 d notice / 3 to 21 days before they ship	`data/bulk_orders.csv` · due_date - placed_date (min, max)	-	✓
F-15	on hand + on order − orders already placed	`src/ddmrp_buffers/buffers.py:99-101`; `policies.py:160-165` · net_flow()	-	✓
F-16	red 105 / yellow 205 / green 275 / net flow 170 -> 140 / Order 144	`tests/test_buffers.py:12-27` (`test_zones_match_hand_worked_example`) · assertions in the test	275 - 140 = 135 -> 144 (cases of 12)	✓
F-17	Without the customer order: 108.	`tests/test_buffers.py:12-27` inputs; `buffers.order_quantity / round_up` · net flow 170 (120 + 50), TOG 275, MOQ 50, case 12	275 - 170 = 105 -> ceil(105 / 12) x 12 = 108	✓
F-18	94.49% vs 94.51%	`outputs/results.json` · policies['DDMRP, no order-spike visibility'] / ['reorder point (tuned)'].year_2.value_fill_rate	x 100, 2 decimals	✓

Assumptions: A-02, A-05

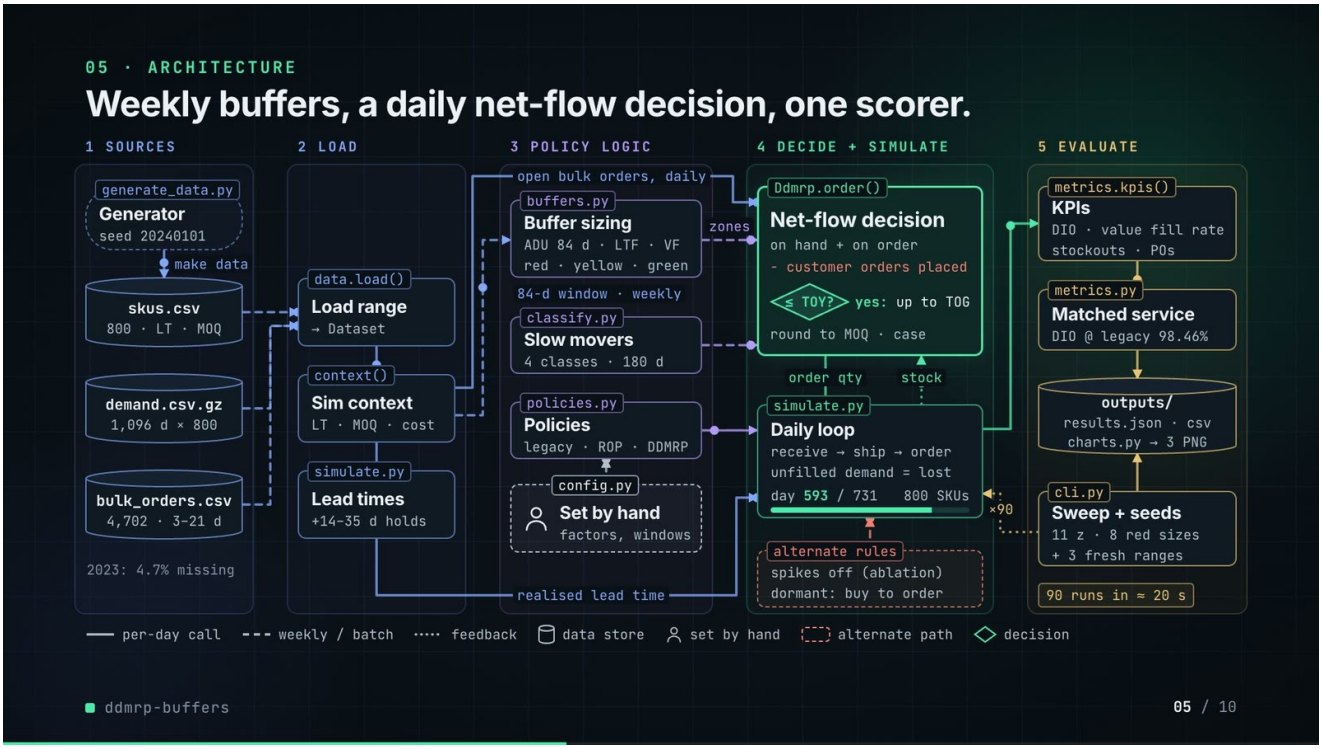
Illustrative, not data:

- The buffer bar is drawn to scale for the hand-worked test SKU (tests/test_buffers.py), not a SKU from the range.

To change this slide: Slide B markup and the 'hand-worked buffer' block in the explainer timeline; the 108 is derived (F-17).

Slide 05: Architecture

26.0-40.8 s · composition `compositions/s03-architecture.html` (node markup + the wires list)





On-screen text (verbatim)

05 · ARCHITECTURE — Weekly buffers, a daily net-flow decision, one scorer.

1 SOURCES: Generator (generate_data.py, seed 20240101) → skus.csv 800 · LT · MOQ · demand.csv.gz 1,096 d × 800 · bulk_orders.csv 4,702 · 3–21 d · 2023: 4.7% missing

2 LOAD: Load range (data.load(), → Dataset) · Sim context (context(), LT · MOQ · cost) · Lead times (simulate.py, +14–35 d holds)

3 POLICY LOGIC: Buffer sizing (buffers.py, ADU 84 d · LTF · VF, red · yellow · green) · Slow movers (classify.py, 4 classes · 180 d) · Policies (policies.py, legacy · ROP · DDMRP) · Set by hand (config.py, factors, windows)

4 DECIDE + SIMULATE: Net-flow decision (Ddmrp.order(): on hand + on order − customer orders placed; ≤ TOY? yes: up to TOG; round to MOQ · case) · Daily loop (simulate.py: receive → ship → order; unfilled demand = lost; day 0→731 / 731, 800 SKUs) · alternate rules (spikes off (ablation); dormant: buy to order)

5 EVALUATE: KPIs (metrics.kpis(): DIO · value fill rate, stockouts · POs) · Matched service (metrics.py: DIO @ legacy 98.46%) · outputs/ (results.json · csv, charts.py → 3 PNG) · Sweep + seeds (cli.py: 11 z · 8 red sizes, + 3 fresh ranges) · 90 runs in ≈ 20 s

Edge labels: make data · open bulk orders, daily · 84-d window · weekly · realised lead time · zones · order qty · stock · ×90

Legend: per-day call · weekly / batch · feedback · data store · set by hand · alternate path · decision

Second phase (38.0-40.8 s): critical path: bulk order → net flow → fill rate (bulk_orders.csv → data.load() → context() → Ddmrp.order() → daily loop → KPIs → matched service → outputs/)

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	800 SKUs	`outputs/results.json` · dataset.skus	-	✓
F-08	4,702 bulk orders	`outputs/results.json; data/bulk_orders.csv` · dataset.bulk_orders	-	✓
F-09	3–21 d notice / 3 to 21 days before they ship	`data/bulk_orders.csv` · due_date - placed_date (min, max)	-	✓
F-11	84-day average / ADU 84 d / 84-d window	`src/ddmrp_buffers/config.py:24` · adu_window_days = 84	-	✓
F-12	EVERY 7 DAYS / weekly	`src/ddmrp_buffers/config.py:25; policies.py:154` · recalc_every_days = 7	-	✓
F-13	EVERY DAY / all 800 SKUs, daily / per-day call	`src/ddmrp_buffers/simulate.py:77-88` · for t in range(n_days): ... policy.order(...)	-	✓
F-19	seed 20240101	`src/ddmrp_buffers/config.py:8` · SEED	-	✓
F-20	1,096 d × 800 / 2023: 4.7% missing	`outputs/results.json` · dataset.days_total, dataset.history_missing_share	-	✓
F-21	+14–35 d holds	`src/ddmrp_buffers/simulate.py:49-50` · 10% of import orders held rng.integers(14, 36) days	-	✓
F-22	ADU 84 d · LTF · VF / red · yellow · green / ≤ TOY? yes: up to TOG	`src/ddmrp_buffers/buffers.py:35-119` · average_daily_usage, lead_time_factor, variability_factor, zones, net_flow, order_quantity	-	✓
F-23	4 classes · 180 d	`src/ddmrp_buffers/classify.py:13-14; config.py:66` · RUNNER, REPEATER, STRANGER, DORMANT; window_days = 180	-	✓
F-24	day 731 / 731 · 731 simulated days	`outputs/results.json` · dataset.days_simulated	-	✓
F-25	11 z · 8 red sizes	`src/ddmrp_buffers/cli.py:14-15` · len(ROP_Z_SWEEP), len(DDMRP_RED_SWEEP)	-	✓
F-26	+ 3 fresh ranges / 4 ranges	`outputs/results.json robustness; cli.py:252` · len(robustness) = 4 (main + 3)	-	✓
F-27	×90 / 90 runs	`src/ddmrp_buffers/cli.py:18-25, 76-93, 120-131, 175-177` · 5 arms + 19 sweep runs on the main range; 3 policies + 19 sweep runs on each of 3 extra ranges	5 + 19 + 3 x (3 + 19) = 90	✓
F-28	≈ 20 s / ~20 s	`README 'Running it'` · '(~20 s)'	-	✓
F-29	98.46%	`outputs/results.json` · matched_service.target_value_fill_rate_year_2	x 100	✓
F-30	generate_data.py, data.load(), context(), simulate.py, buffers.py, classify.py, policies.py, config.py, Ddmrp.order(), metrics.kpis(), metrics.py, cli.py, charts.py, outputs/	`src/ddmrp_buffers/, scripts/` · files + functions	-	✓

F-28 note: Stated in the README; results.json does not record a run time, and it depends on the machine.

Assumptions: A-01, A-02, A-04, A-05, A-07, A-08

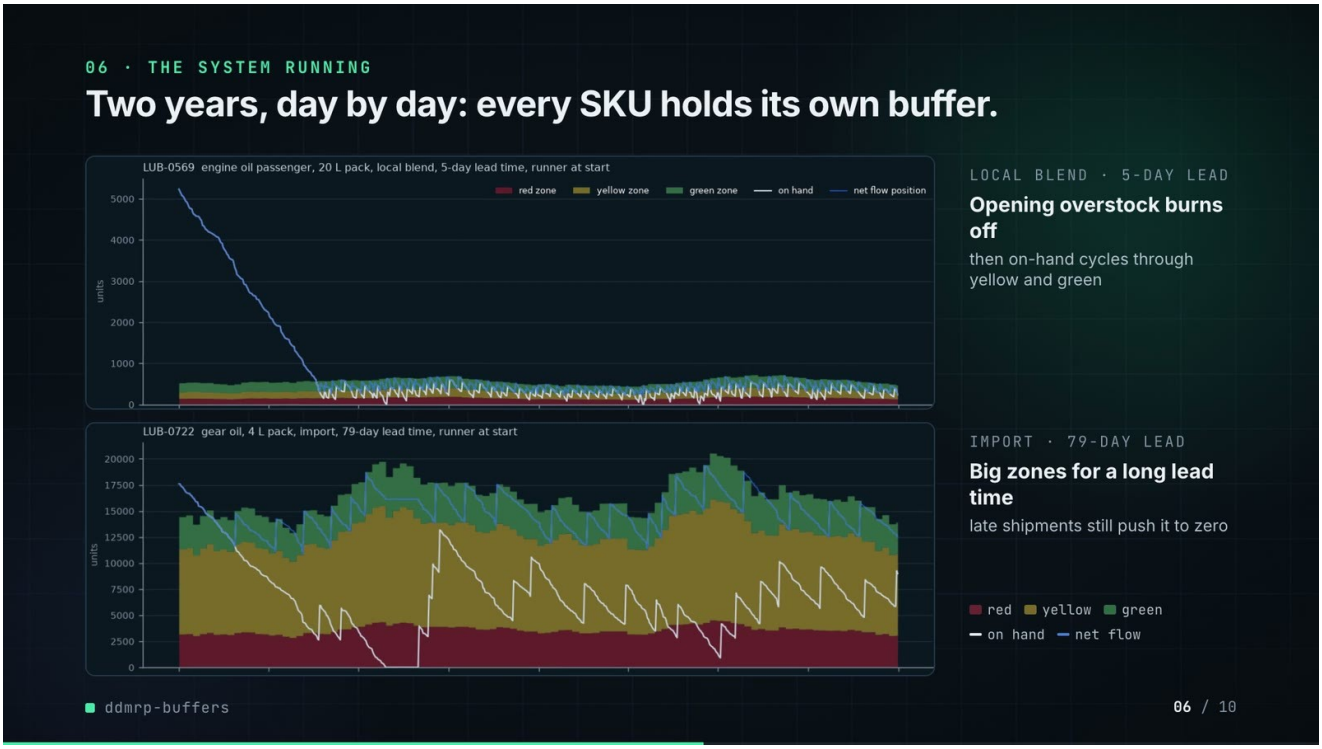
Illustrative, not data:

- Packets and the day counter show execution order, not real-time speed.
- 'alternate rules' groups two real code paths: the no-spike ablation arm (use_spikes=False) and the slow-mover rule (dormant SKUs bought only against an order in hand).
- The critical-path highlight is emphasis only.

To change this slide: Boxes are absolutely positioned divs in compositions/s03-architecture.html; wires are the `wires` point lists in its script (keep them orthogonal and matching the box edges); `crit: true` and `critNodes` set the highlighted path.

Slide 06: The system running

40.4-55.0 s · composition `compositions/s04-routes.html`





On-screen text (verbatim)

06 · THE SYSTEM RUNNING — Two years, day by day: every SKU holds its own buffer.
LOCAL BLEND · 5-DAY LEAD — Opening overstock burns off, then on-hand cycles through yellow and green
IMPORT · 79-DAY LEAD — Big zones for a long lead time, late shipments still push it to zero
Second phase: Same fill rate as today, far less stock.
AT TODAY'S FILL RATE · 98.46% — days of stock each policy needs (year 2): Min-max (legacy) 97.6 · Reorder point (tuned) 65.9 · DDMRP 52.1 — 52.1 days −47%
each policy swept over its one safety knob, then read at the legacy fill rate

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-29	98.46%	outputs/results.json · matched_service.target_value_fill_rate_year_2	x 100	✓
F-31	LOCAL BLEND · 5-DAY LEAD / IMPORT · 79-DAY LEAD	outputs/buffer_chart.png (panel titles); data/skus.csv · sku, source, lead_time_days	-	✓
F-32	legacy fill 98.46% + three rings	outputs/frontier.png (axes); compositions/s04-routes.html (X/Y mapping) · tick positions in the dark image	x = 332 + (fill - 90) x 150.1, y = 968 - (DIO - 30) x 13.514 (image pixels)	manual
F-33	97.6 / 65.9 / 52.1	outputs/results.json · policies['min-max (legacy)'].year_2.dio_days; matched_service.dio_at_target	-	✓
F-34	–47%	outputs/results.json · F-33	(52.14 / 97.64 - 1) x 100 = -46.6%	✓

Assumptions: A-04, A-07, A-08, A-09

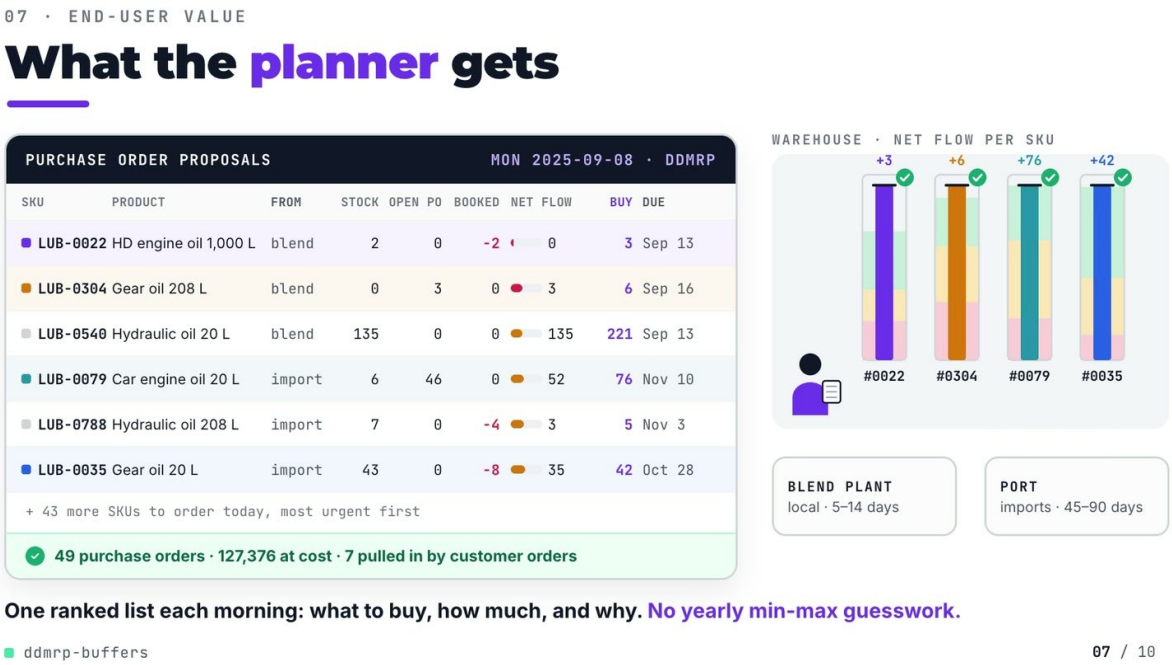
Illustrative, not data:

- Panels are outputs/buffer_chart.png and outputs/frontier.png converted to dark by tools/darkmap.py; nothing is redrawn.
- The dashed line and rings on the frontier are annotations (F-32).
- Side captions describe what the two plotted SKUs show (README 'How it works').

To change this slide: Re-run tools/darkmap.py after make run; bar widths in #routes-f1..3 are 520 x DIO / 97.6; the counter tweens 97.6 → 52.1.

Slide 07: End-user value (white)

54.7-67.7 s · composition `compositions/value.html` (SHEET injected by `tools/inject_sheet.py`)



On-screen text (verbatim)

07 · END-USER VALUE — What the planner gets

PURCHASE ORDER PROPOSALS · MON 2025-09-08 · DDMRP

SKU · PRODUCT · FROM · STOCK · OPEN PO · BOOKED · NET FLOW · BUY · DUE

LUB-0022 HD engine oil 1,000 L blend 2 0 −2 0 3 Sep 13

LUB-0304 Gear oil 208 L blend 0 3 0 3 6 Sep 16

LUB-0540 Hydraulic oil 20 L blend 135 0 0 135 221 Sep 13

LUB-0079 Car engine oil 20 L import 6 46 0 52 76 Nov 10

LUB-0788 Hydraulic oil 208 L import 7 0 −4 3 5 Nov 3

LUB-0035 Gear oil 20 L import 43 0 −8 35 42 Oct 28

+ 43 more SKUs to order today, most urgent first

49 purchase orders · 127,376 at cost · 7 pulled in by customer orders

Scene: WAREHOUSE · NET FLOW PER SKU (#0022 +3, #0304 +6, #0079 +76, #0035 +42) · BLEND PLANT local · 5–14 days · PORT imports · 45–90 days

One ranked list each morning: what to buy, how much, and why. No yearly min-max guesswork.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-35	MON 2025-09-08 · DDMRP	`demo assets/po_sheet.json` (re-run of the repo's default DDMRP simulation) · day, weekday	-	✓
F-36	49 purchase orders	`demo assets/po_sheet.json` · orders = number of SKUs with order qty > 0	-	✓
F-37	127,376 at cost	`demo assets/po_sheet.json` · order_value = sum(order qty x unit cost)	-	✓
F-38	7 pulled in by customer orders	`demo assets/po_sheet.json` · with_customer_orders = 7; triggered_by_customer_orders = 7 (net flow without them would sit above top of yellow)	-	✓
F-39	+ 43 more SKUs to order today	F-36 · 49 - 6 rows	49 - 6 = 43	✓
F-40	LUB-0022, LUB-0304, LUB-0540, LUB-0079, LUB-0788, LUB-0035	`demo assets/po_sheet.json, injected into compositions/value.html` · rows by SKU	due = order day + nominal lead time; rows ordered most urgent first (net flow / top of green); LUB-0540 kept as a high-volume line	✓
F-41	local · 5–14 days / imports · 45–90 days	`data/skus.csv` · lead_time_days by source (min, max)	-	✓
F-42	+3 / +6 / +76 / +42 with green checks	`demo assets/po_sheet.json` · net_flow + order_qty >= tog	-	✓

Assumptions: A-04, A-05, A-11

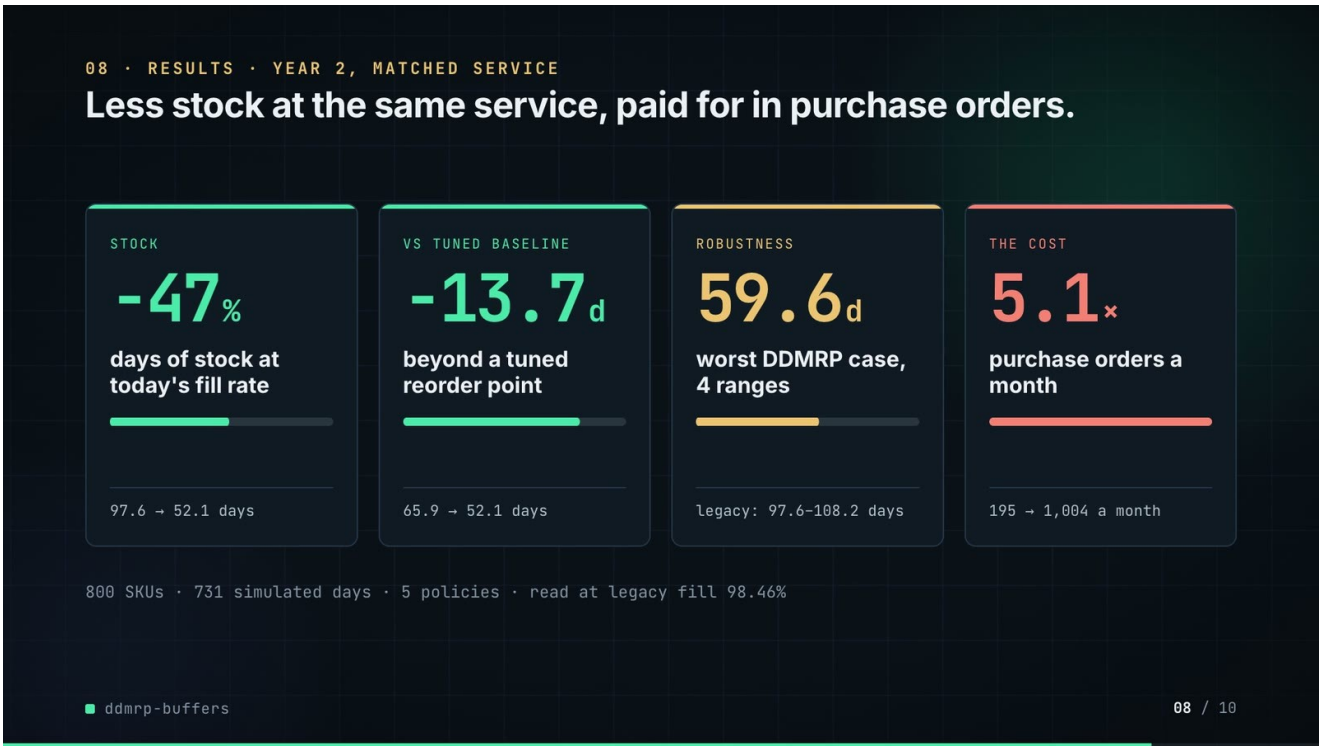
Illustrative, not data:

- The planner, trucks and supplier boxes are a flat 2D scene, not data.
- Gauge zones and levels are to scale per SKU from po_sheet.json (red / yellow / green, net flow, net flow + order); each gauge has its own scale.
- DUE = order day + nominal lead time; the simulation's realised lead time can differ (A-04).
- 'Booked' = customer orders already placed that count against the buffer (qualified spikes).

To change this slide: Change the day in tools/export_po_sheet.py, re-run it and tools/inject_sheet.py (choose --rows / --animate from the new JSON), then render.

Slide 08: Results

67.4-76.2 s · composition `compositions/s05-results.html`



On-screen text (verbatim)

08 · RESULTS · YEAR 2, MATCHED SERVICE — Less stock at the same service, paid for in purchase orders.
STOCK −47% days of stock at today's fill rate · 97.6 → 52.1 days
VS TUNED BASELINE −13.7 d beyond a tuned reorder point · 65.9 → 52.1 days
ROBUSTNESS 59.6 d worst DDMRP case, 4 ranges · legacy: 97.6–108.2 days
THE COST 5.1× purchase orders a month · 195 → 1,004 a month
800 SKUs · 731 simulated days · 5 policies · read at legacy fill 98.46%

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	800 SKUs	`outputs/results.json` · dataset.skus	-	✓
F-24	day 731 / 731 · 731 simulated days	`outputs/results.json` · dataset.days_simulated	-	✓
F-26	+ 3 fresh ranges / 4 ranges	`outputs/results.json robustness; cli.py:252` · len(robustness) = 4 (main + 3)	-	✓
F-29	98.46%	`outputs/results.json` · matched_service.target_value_fill_rate_year_2	x 100	✓
F-34	−47%	`outputs/results.json` · F-33	(52.14 / 97.64 - 1) x 100 = -46.6%	✓
F-43	−13.7 d (65.9 → 52.1 days)	`outputs/results.json` · matched_service.dio_at_target	65.888 - 52.140 = 13.75 -> 13.7	✓
F-44	59.6 d	`outputs/results.json` · max(robustness[].ddmrp_dio_at_legacy_fill)	max over 4 ranges	✓
F-45	legacy: 97.6–108.2 days	`outputs/results.json` · robustness[].legacy_dio (min, max)	-	✓
F-46	5.1×	`outputs/results.json` · policies[*].year_2.purchase_orders_per_month: 1,004.1 / 195.1	1,004.1 / 195.1 = 5.15	✓
F-47	195 → 1,004 a month	`outputs/results.json` · policies[*].year_2.purchase_orders_per_month	-	✓
F-48	5 policies	`outputs/results.json` · len(policies)	-	✓

F-43 note: The README rounds the inputs first (65.9 - 52.1 = 13.8 days); the slide shows the difference of the unrounded values.

Assumptions: A-07, A-08, A-09, A-12

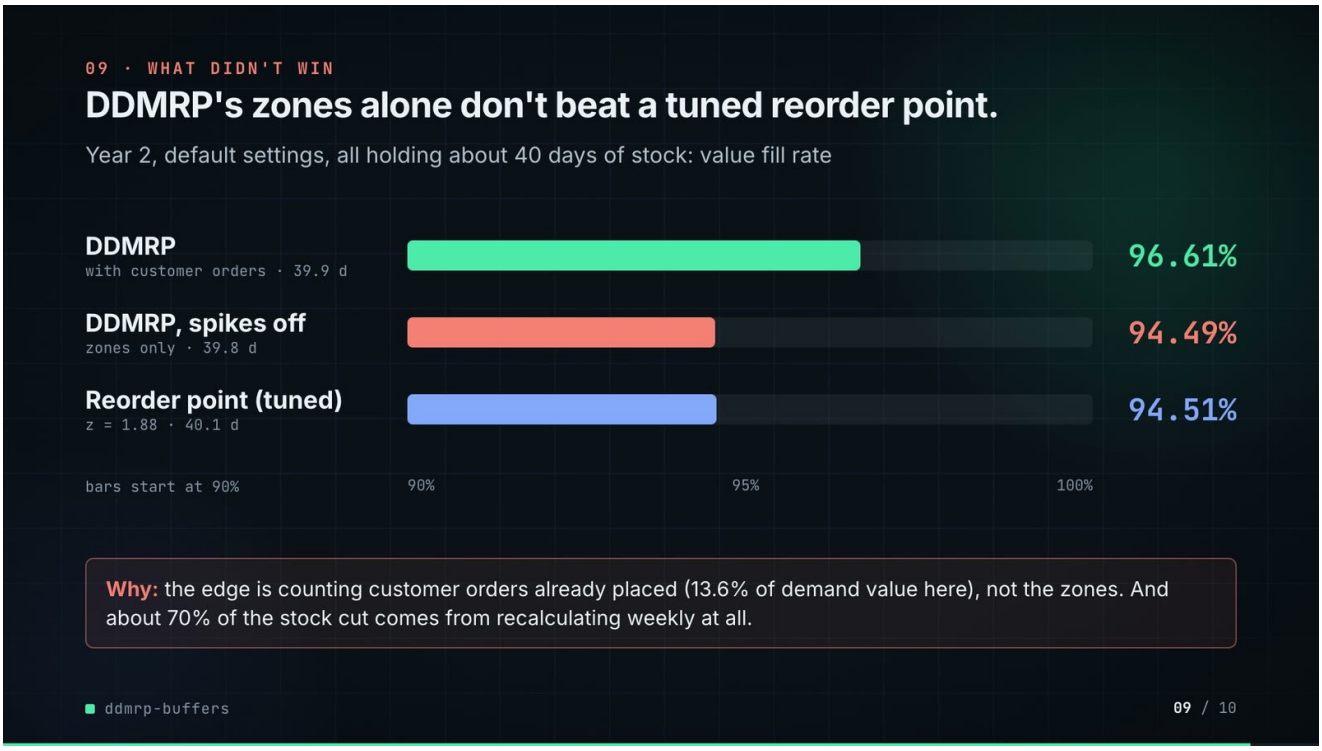
Illustrative, not data:

- Mini bars are proportional to the shown ratios.

To change this slide: Edit the `stats` array and card text in compositions/s05-results.html.

Slide 09: What didn't win

75.8-82.4 s · composition `compositions/s06-twist.html`



On-screen text (verbatim)

09 · WHAT DIDN'T WIN — DDMRP's zones alone don't beat a tuned reorder point.

Year 2, default settings, all holding about 40 days of stock: value fill rate

DDMRP (with customer orders · 39.9 d) 96.61% · DDMRP, spikes off (zones only · 39.8 d) 94.49% · Reorder point (tuned) (z = 1.88 · 40.1 d) 94.51% · bars start at 90%

Why: the edge is counting customer orders already placed (13.6% of demand value here), not the zones. And about 70% of the stock cut comes from recalculating weekly at all.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-18	94.49% vs 94.51%	`outputs/results.json` · policies['DDMRP, no order-spike visibility'] / ['reorder point (tuned)'].year_2.value_fill_rate	x 100, 2 decimals	✓
F-49	96.61% · 39.9 d	`outputs/results.json` · policies.DDMRP.year_2	-	✓
F-50	94.49% · 39.8 d	`outputs/results.json` · policies['DDMRP, no order-spike visibility'].year_2	-	✓
F-51	94.51% · 40.1 d	`outputs/results.json; config.py:59` · policies['reorder point (tuned)'].year_2	-	✓
F-52	13.6% of demand value	`outputs/results.json` · dataset.bulk_share_of_simulated_demand_value = 0.13556	x 100	✓
F-53	about 70%	`outputs/results.json` · legacy DIO, matched-service DIO of reorder point and DDMRP	(97.64 - 65.89) / (97.64 - 52.14) = 0.698	✓

Assumptions: A-02, A-05, A-07

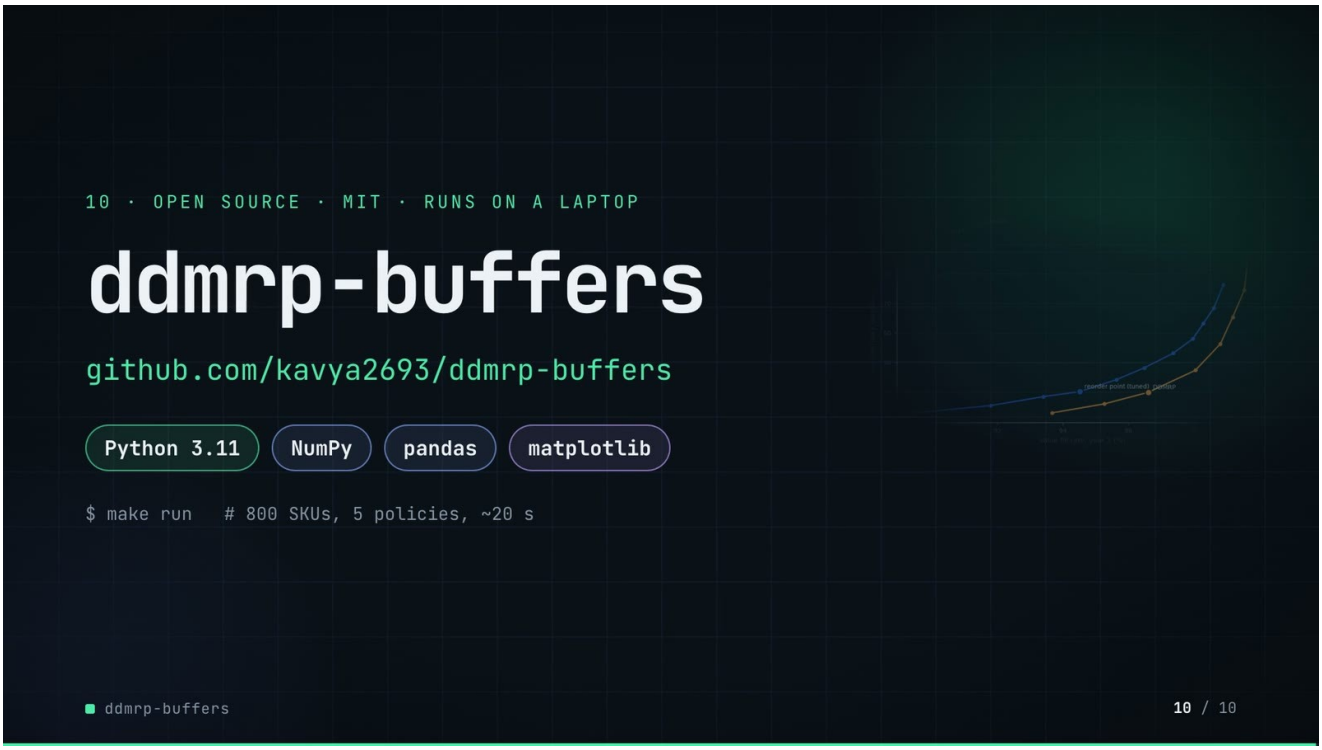
Illustrative, not data:

- Bar width = (fill - 90%) x 100 px per point.

To change this slide: Rows #twist-r1..r3 and #twist-verdict in compositions/s06-twist.html.

Slide 10: End card

82.0-86.4 s · composition `compositions/s07-end.html`



On-screen text (verbatim)

10 · OPEN SOURCE · MIT · RUNS ON A LAPTOP
ddmrp-buffers
github.com/kavya2693/ddmrp-buffers
Python 3.11 · NumPy · pandas · matplotlib
$ make run # 800 SKUs, 5 policies, ~20 s

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	800 SKUs	`outputs/results.json` · dataset.skus	-	✓
F-28	≈ 20 s / ~20 s	`README 'Running it'` · '(~20 s)'	-	✓
F-48	5 policies	`outputs/results.json` · len(policies)	-	✓
F-54	MIT	`LICENSE line 1` · -	-	✓
F-55	Python 3.11 · NumPy · pandas · matplotlib	`pyproject.toml` · requires-python, dependencies	-	✓
F-56	$ make run	`Makefile` · run target	-	✓

F-28 note: Stated in the README; results.json does not record a run time, and it depends on the machine.

Assumptions: A-13

Illustrative, not data:

- Faded chart in the background is outputs/frontier.png.

To change this slide: Text in compositions/s07-end.html.

5. Change-impact matrix

Input / assumption	Affected figures	Slides	Files to edit	Re-run
A-01: All data is synthetic, produced by a seeded generator sized to the production problem (800 SKUs, three years of daily demand, 4,702 bulk orders)	all figures	01, 02, 03, 04, 05, 06, 07, 08, 09, 10	`src/ddmrp_buffers/data.py`; `src/ddmrp_buffers/config.py`	make data; make run
A-02: Bulk fleet orders are announced 3 to 21 days before they ship: 60% of SKUs packed in 18 L or more get 2-8 such orders a year (15% of the rest get 2)	F-08, F-09, F-16, F-17, F-18, F-36, F-37, F-38, F-49, F-50, F-51, F-52	02, 04, 05, 07, 09	`src/ddmrp_buffers/data.py:157-169`	make data; make run; export_po_sheet.py + inject_sheet.py
A-03: Unfilled demand is lost, not backordered (a workshop buys the drum elsewhere)	F-18, F-29, F-33, F-34, F-43, F-44, F-49, F-50, F-51, F-53	04, 05, 06, 08, 09	`src/ddmrp_buffers/simulate.py:81-83`	make run
A-04: Lead times: local blends 5-14 days, imports 45-90 days	F-07, F-21, F-33, F-34, F-41, F-44, F-49, F-50, F-51	02, 05, 06, 07, 08, 09	`src/ddmrp_buffers/data.py:109-114`; `src/ddmrp_buffers/simulate.py:46-50`	make data; make run
A-05: DDMRP settings: 84-day ADU window, buffers recalculated every 7 days, lead-time factor 0	F-11, F-12, F-22, F-33, F-34, F-36, F-37, F-38, F-40, F-42, F-43, F-44, F-49, F-50	03, 05, 06, 07, 08, 09	`src/ddmrp_buffers/config.py:24-41`	make run; export_po_sheet.py + inject_sheet.py
A-06: Legacy min-max: min = usage over lead time + 30 days, max = min + 90 days (blend) or 180 days (import), reset once a year from last year's average daily usage	F-29, F-33, F-34, F-45, F-47, F-53	05, 06, 08, 09	`src/ddmrp_buffers/config.py:48-50`	make run
A-07: Tuned reorder point: z = 1	F-25, F-27, F-33, F-34, F-43, F-44, F-51, F-53	05, 06, 08, 09	`src/ddmrp_buffers/config.py:59`; `src/ddmrp_buffers/cli.py:14-15`	make run
A-08: Stock at matched service = DIO linearly interpolated along each policy's swept (fill rate, DIO) curve at the legacy policy's year-2 fill rate (98	F-14, F-29, F-33, F-34, F-43, F-44, F-53	03, 05, 06, 08, 09	`src/ddmrp_buffers/metrics.py:42-49`	make run
A-09: Headline window is year 2 of the simulation (2025), after the opening overstock has worked through; 2023 is history only, 2024-2025 are simulated (731 days)	F-24, F-29, F-33, F-34, F-43, F-44, F-45, F-46, F-47, F-49, F-50, F-51, F-53	05, 06, 08, 09	`src/ddmrp_buffers/config.py:10-12`	make data; make run
A-10: Opening stock reflects generous cover (lognormal around 110 days, x1	F-04, F-05	02	`src/ddmrp_buffers/data.py:182-184`	make data; make run
A-11: Lead-time draws use simulation seed 7 for the main range; the three extra ranges use their own seed for both the range and the draws	F-35, F-36, F-37, F-38, F-39, F-40, F-42, F-44, F-45	07, 08	`docs/demo/source/tools/export_po_sheet.py (day argument)`	export_po_sheet.py + inject_sheet.py; hyperframes render
A-12: Purchase orders are counted, not costed: there is no ordering or freight cost in the comparison	F-46, F-47	08	`src/ddmrp_buffers/metrics.py`	make run
A-13: Video rounding: days to 1 decimal, fill rates to 2 decimals, headline percentages to whole numbers, money to 0	F-04, F-05, F-06, F-10, F-14, F-20, F-34, F-43, F-46, F-52, F-53	02, 03, 05, 06, 08, 09	`the composition file named on each slide`	hyperframes render
A-14: Slide 01 (industries, everyday problems) is general distribution knowledge used as context	F-01, F-02	01	`compositions/s01-context.html`	hyperframes render

6. Known differences and caveats

• The data is generated (A-01). DDMRP's edge over the reorder point comes from bulk orders being visible 3-21 days ahead; that share and notice period are generator choices (A-02), so with a real order book the gap could be larger or close to zero (README 'Known limitations').

• '47% less stock' and 'same fill rate' hold at matched service (DIO interpolated at the legacy fill rate, A-08). At default settings DDMRP's fill rate is lower than legacy (96.61% vs 98.46%) and lost sales more than double.

• −13.7 days (slide 08) is the difference of the unrounded matched-service DIOs (13.75); the README's 13.8 subtracts the rounded values (65.9 − 52.1).

• 'Worst DDMRP case' (59.6 d) and the legacy range (97.6-108.2 d) are over the main range and three freshly generated ones; on one of them (seed 20240103) the tuned reorder point never reaches the legacy fill rate in its sweep.

• Slide 07 shows one day (2025-09-08) re-run by tools/export_po_sheet.py with the repo's own DDMRP policy, data and seed; the day was chosen by the video author. DUE dates use the nominal lead time.

- Purchase orders are counted, not costed (A-12): the 5.1x workload is real in the simulation, its cost is not modelled.
- The architecture slide groups two code paths as 'alternate rules': the no-spike ablation arm and the slow-mover rule (strangers keep the red zone only, dormant SKUs are bought only against an order in hand).
- Slide 01 statements are general industry knowledge (A-14).
- make run time (~20 s) is the README's statement for the author's laptop; it is not recorded in results.json.

Appendix: environment

Item	Value
Commit	`ef2e4977bbb5278cf24e612031c686ff3d296161`
Python	3.11.7
numpy	2.4.6
pandas	3.0.6
matplotlib	3.11.2
Profile / seeds	data_seed=20240101, lead_time_draw_seed=7, extra_range_seeds=20240102, 20240103, 20240104
Run time	README 'Running it': make run takes about 20 s on the author's laptop (results.json does not record a run time)